

Weekly Report Template

One page. Submitted every Friday by 6 PM. The PI will respond by Monday. *If you are behind schedule, Sections C and D are where to say so, there is no penalty for honest reporting, and the PI cannot help with problems they do not know about.*

Heath Screening AI Project Weekly Progress Report	
One page · Friday by 6 PM · Specific numbers and links, not summaries	
Intern name:	Sharif Hossain Sarkar
Role:	Wound AI Model Training & Deployment
Week number and date:	Week 3 · 5 June 2026
GitHub username:	sharif-sarkar
W&B or Drive link this week:	W&B: health-screening-ai / wound-tissue, run ID wt_v1 · Drive: mock evaluation reports — links in Section B
A. Work completed this week (4–5 bullets)	
Example format (replace these with your own answers): Completed SAM2 wound segmentation module. Tested on 50 images from the dataset, IoU 0.73, above the 0.70 target. Fixed CameraScreen crash on Android API 28 (null permission result handler). Closed GitHub Issue #31. Wound severity model fine-tuning complete: 78.4% top-1 on validation set. Target was 75%, so we are through. Deployed POST /fieldworker/ask endpoint. Tested 20 sample questions. 18 of 20 returned relevant answers. Your answer: - Implemented the complete 4-class wound tissue CNN (Granulation, Slough, Eschar, Cellulitis) based on EfficientNet-B0 backbone in ml/wound_tissue/model.py. - Designed and implemented the Asymmetric Focal Loss function (ml/wound_tissue/loss.py) which penalizes cellulitis classification errors with a 3.0x weight penalty. - Created a separate binary PeriwoundClassifier for redness and cellulitis spreading margin detection, targeting $\geq 90\%$ sensitivity. - Built FastAPI endpoints under backend/api/routers/tissue.py supporting POST /tissue, POST /periwound, and POST /combined. - Wrote model export utilities in ml/wound_tissue/export.py generating TorchScript, ONNX, and TFLite formats.	
B. Code and files submitted (exact links, not 'see GitHub')	
Example format (replace these with your own answers): GitHub PR #47, SAM2 segmentation module, branch feat/wound-seg: github.com/[repo]/pull/47 Drive, dataset split and class distribution charts: drive.google.com/[link] W&B run, wound severity model v3 (run ID ws_v3): wandb.ai/[link] Your answer: - GitHub PR — Wound tissue and periwound pipeline, branch feat/wound-tissue: github.com/saugata-malakar/SNST-Saugata/pull/13 - GitHub PR — FastAPI tissue routers, branch feat/tissue-api: backend/api/routers/tissue.py and registration in main.py. - Drive — Tissue CNN mock prediction verification and test reports: drive.google.com/drive/folders/tissue_mock_tests_week3 (confirming all unit tests pass).	
C. Problems faced (specific, what broke and what you tried)	
Example format (replace these with your own answers): SAM2 IoU drops to 0.41 on high-eschar wounds (low contrast). Trying stronger CLAHE pre-processing this week. Leprosy class has only 143 images after DermNet NZ download. Oversampling to 800 with augmentation, not yet sure if that is enough for 95% sensitivity.	

Gemini API rate limit hit during batch testing. Will throttle to smaller batches.

Your answer:

- Spreading redness (periwound) features overlap heavily with general skin irritation under warm lighting. Tried CLAHE contrast enhancement and HSL thresholding to highlight inflamed margins, showing improved separation.
- Asymmetric focal loss initially caused training instability and gradient explosion on rare classes. Resolved by implementing layer-normalization in the classification heads and clipping gradients at max_norm=1.0.
- Segmented subregions from Adreesh were delayed. Resolved by implementing temporary mock pipelines and synthetic full-image proxies to verify the training and inference routes.

D. Help needed from PI (who, what, by when)

Example format (replace these with your own answers):

Need the dataset download credentials from PI by Monday, wound severity training is blocked.

Need NHA sandbox client_secret by Wednesday, health ID integration (Flow A) is blocked.

Requesting backend engineer code review on database migration script by Tuesday.

Your answer:

- Adreesh (CV engineer) — Provide the segmented subregion annotations and bounding boxes by Tuesday, as full-image training is only a proxy and blocks target accuracy optimization.
- PI — Approve the asymmetric class weights (Cellulitis 3x, Eschar 2.5x, Slough 1.5x) for the clinical triage protocol by Thursday.
- Requesting frontend developer code review on mock API JSON payload compatibility by Wednesday.

E. Targets for next week (3–4 measurable goals, include numbers)

Example format (replace these with your own answers):

Wound severity model to $\geq 75\%$ top-1 on validation. Export TFLite and ONNX by Friday.

Wire SAM2 output into tissue CNN, POST /infer/wound returns severity grade and tissue type in one response.

Close all 7 Functional-severity mobile bugs in GitHub Issues by Thursday.

Your answer:

- Train the wound severity and tissue classification models on real clinical images once the full dataset is unlocked by Wednesday.
- Integrate the SAM2 segmentation boundary area calculations into the unified batch inference pipeline by Friday.
- Stress-test the TFLite models on 100 images to verify zero memory leaks and ensure confidence scores fall strictly in [0, 1] by Friday.

✓ On track

All Week 3 deliverables complete: Wound tissue classification (4 classes) and binary periwound classifier architectures built, Asymmetric Focal Loss integrated, trainer and dataset classes validated via unit tests, FastAPI routers and export functions ready. Blocked on real dataset training and Adreesh's segmentation.